Registration No.:_____

PNR No:: DEC507659

COURSE CODE : DECAP790
COURSE NAME : PROBABILITY AND STATISTICS

Time Allowed: 03:00 hrs ***Max.Marks: 80***

1. This question paper is divided into two parts A and B.
2. Answer all the questions in serial order.
3. Part A contains 10 questions of 2 marks each. All questions are compulsory.
4. Part B contains 10 questions (Questions 2 to 11) of 10 marks each, attempt any 06 questions out of 10. Attempt all parts of the selected question. Only first 06 attempted questions would be evaluated.
5. The student is required to attempt the question paper in English medium only.
6. Simple non programmable calculator is allowed.

PART A

Q1(a) When a die is rolled, what is the likelihood of receiving a 2 or a 5?
(b) What is random sampling ?Explain with example .
(c) What is skewness ?Why it is important
(d) Which measure of central tendency is better if many outliers are present in data ?
(e) What are raw moments in statistics?
(f) What types of Research issues can regression analysis answer?
(g) Why are X bar and R charts used together?
(h) How time series analysis helps organizations understand the underlying causes of trends or systemic patterns over time?
(i) Write example for alternate hypothesis?
(j) What is p value in ANOVA?

PART B

Q2 What is sampling design and explain with example?
Q3 How Dispersion is measured? Explain it with example
Q4 What are different measures of location explain with example of each
Q5 What is T-test used for, explain it with an example.
Q6 In what scenario p and Np chart is used?
Q7 What are the major features of SPSS?
Q8 Why skewness is a measure of the asymmetry of the probability distribution of a random variable about its mean?
Q9 What are different methods to measure correlation and regression?
Q10 What is mean, median and mode ?Why mean median and mode of perfect normal distribution is Equal ?
Q11 Explain the concept of normal distribution. Discuss some examples of normal distribution.

-- End of Question Paper --